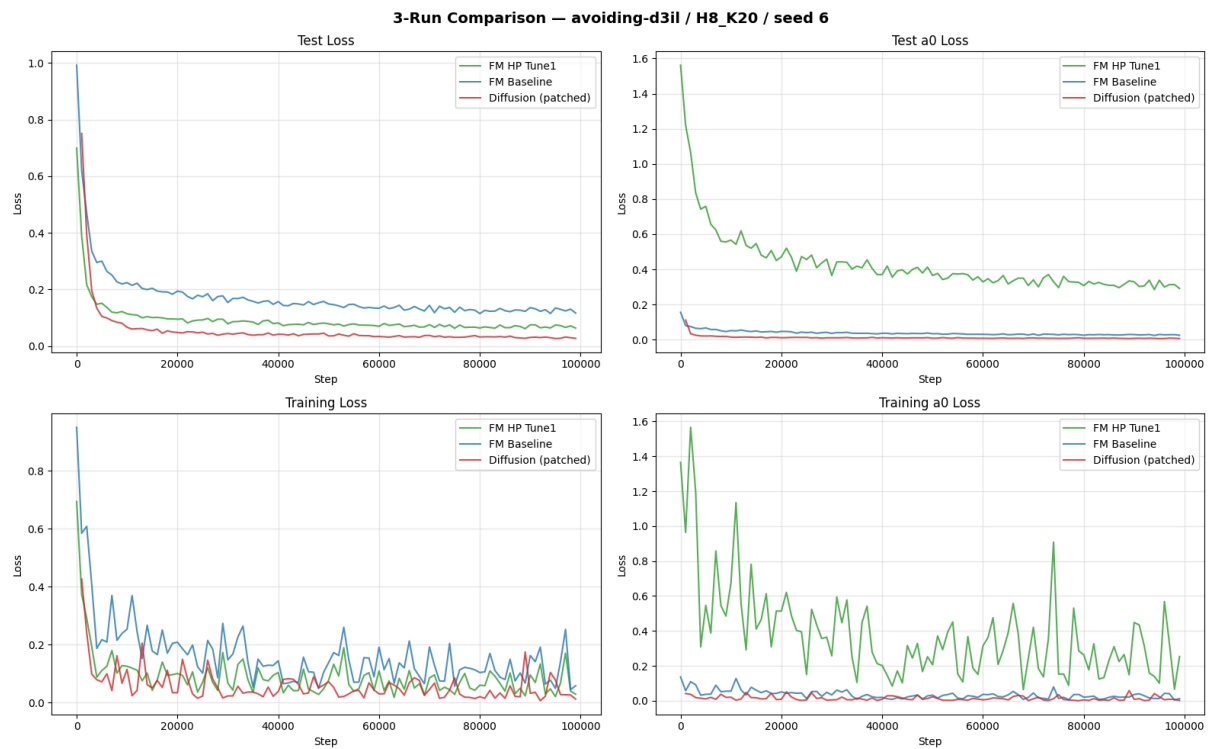
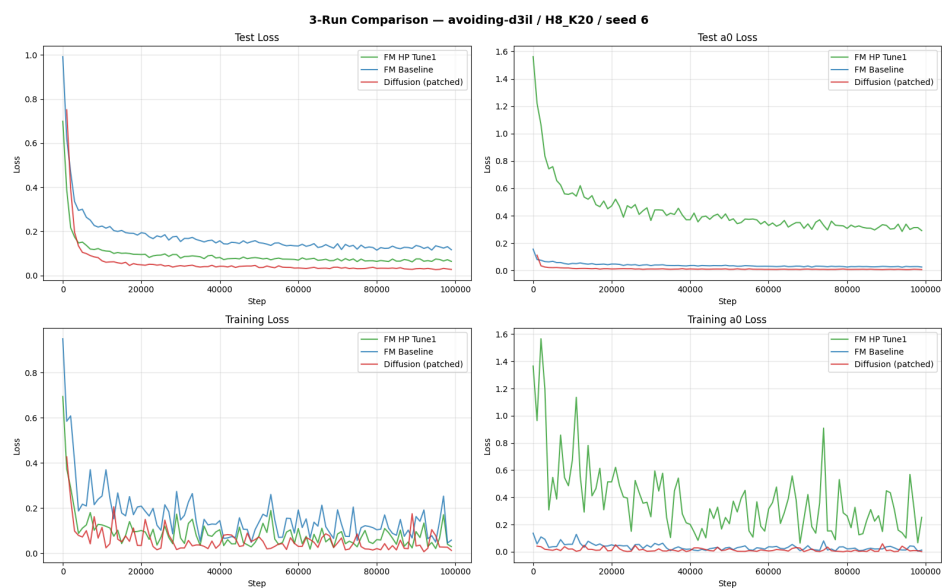


Training_Losses_pkl_analysis_Seed6_HP1_vs_F



From Train_Curve_Analysis_Seed_6_HP1_vs_FM_vs_D_Colab_GPU_FMPCC_MuJoCo_boosted.ipynb
in
\\load_results_outputs\Data_Analysis\Training_Losses_pkl_analysis_Seed6_HP1_vs_FM_vs_D
or
\\ipynbs\Test_Results_from_Colab\Train\HP1\Data_Analysis_Seed6_HP1_vs_FM_vs_D



```
=====
Run                      Final Test Loss    Min Test Loss    Final a0 Test    Min a0 Test
=====
FM HP Tune1              0.063971        0.061796        0.291627        0.285329
FM Baseline              0.116854        0.115070        0.025723        0.024426
Diffusion (patched)     0.027634        0.026863        0.007069        0.006796
=====
```

Convergence speed (first step where test_loss < threshold):

```
threshold = 0.2
FM HP Tune1      : step 3000
FM Baseline      : step 14000
Diffusion (patched) : step 2999
```

```
threshold = 0.15
FM HP Tune1      : step 4000
FM Baseline      : step 39000
Diffusion (patched) : step 3999
```

```
threshold = 0.12
FM HP Tune1      : step 7000
FM Baseline      : step 71000
Diffusion (patched) : step 4999
```

Overfitting gap at final step (train_loss - test_loss):

```
FM HP Tune1      : -0.035390 (overfit)
FM Baseline      : -0.059423 (overfit)
Diffusion (patched) : -0.015783 (overfit)
```

Data Analysis (Seed 6 — HP1 vs FM Baseline vs Diffusion)

1. Final Performance Ranking

Metric	🏆 Best	🥈 Second	🥉 Third
Final Test Loss	Diffusion (0.0276)	FM HP Tune1 (0.0640)	FM Baseline (0.1169)
Min Test Loss	Diffusion (0.0269)	FM HP Tune1 (0.0618)	FM Baseline (0.1151)
Final a0 Test	Diffusion (0.0071)	FM Baseline (0.0257)	FM HP Tune1 (0.2916)
Min a0 Test	Diffusion (0.0068)	FM Baseline (0.0244)	FM HP Tune1 (0.2853)

- Diffusion (patched) ranks first across all four metrics.
- For test loss: Diffusion < HP Tune1 (2.3× gap) < FM Baseline (4.2× gap).
- For a0 test: Diffusion < FM Baseline (3.6× gap) < HP Tune1 (41× gap). Note the **ranking inversion** — HP Tune1 beats Baseline on test loss but is far worse on a0.

2. Test Loss Trajectory

Step	Diffusion	FM HP Tune1	FM Baseline
0–1k	0.751	0.699	0.992
5k	0.0995	0.151	0.300
10k	0.0666	0.114	0.224
25k	0.0493	0.093	0.175

Step	Diffusion	FM HP Tune1	FM Baseline
50k	0.0361	0.079	0.149
75k	0.0311	0.063	0.119
99k	0.0276	0.064	0.117

- Diffusion: 0.75 → 0.10 in 5k steps (7.5× drop), then slow refinement. Plateaus after ~25k.
- HP Tune1: rapid descent 0–10k, slow steady improvement 10–90k. Final 0.064 ≈ min 0.062 — marginal improvement still possible.
- FM Baseline: slowest in every phase. At 50k it's at 0.149, where Diffusion already sat at 0.036.

Convergence Speed

Threshold	HP Tune1	Baseline	Diffusion	BL → HP1	BL → Diff
0.20	3,000	14,000	2,999	4.7×	4.7×
0.15	4,000	39,000	3,999	9.8×	9.8×
0.12	7,000	71,000	4,999	10.1×	14.2×

- HP Tune1 and Diffusion converge at nearly identical speeds for 0.20 and 0.15 thresholds.
- At 0.12, Diffusion is 1.4× faster than HP Tune1.
- FM Baseline is 10–14× slower than both.

Δ Test Loss per Phase

Phase	Diffusion	HP Tune1	Baseline
0–10k	−0.685	−0.585	−0.768
10k–30k	−0.022	−0.036	−0.070
30k–50k	−0.008	0.000	−0.005
50k–75k	−0.005	−0.016	−0.030
75k–99k	−0.003	−0.001	−0.003

- All models do 85%+ of learning in the first 10k steps.
- HP Tune1 has a **stall at 30–50k** ($\Delta = 0.000$), then resumes improving 50–75k ($\Delta = -0.016$).
- Baseline still improving meaningfully at 50–75k ($\Delta = -0.030$), most room for continued training.
- After 75k, all three have marginal returns ($\Delta < 0.003$).

3. Training Noise & Test Stability

Run	Train Range (80k–99k)	Test Range (80k–99k)	Train Spread
Diffusion	0.006 – 0.174	0.027 – 0.035	29×
FM HP Tune1	0.020 – 0.170	0.062 – 0.075	8.5×
FM Baseline	0.049 – 0.252	0.115 – 0.136	5.1×

- All runs: noisy training losses, smooth test losses.
- Diffusion: highest batch-level noise (29×) yet tightest test range (± 0.004).
- Baseline: lowest batch noise (5.1×) but widest test fluctuation (± 0.010).

4. Generalization Gap

Run	train_loss – test_loss (final)	Gap as % of test_loss
Diffusion	−0.0158	57%
FM HP Tune1	−0.0354	55%
FM Baseline	−0.0594	51%

- All runs: train < test (negative gap).

- Absolute gap: Diffusion smallest (0.016), Baseline largest (0.059).
- Relative gap (as % of test loss): surprisingly similar across all three (~51–57%), meaning the overfit ratio is comparable — the absolute gap difference is mainly driven by the difference in loss scale.

5. a0 Test Loss — Data Observations

Observed Values Over Training

Step	Diffusion a0_test	FM Baseline a0_test	HP Tune1 a0_test	HP1 / BL ratio
0–1k	0.112	0.156	1.562	10.0×
10k	0.0155	0.052	0.567	10.9×
25k	0.0135	0.041	0.456	11.1×
50k	0.0087	0.034	0.365	10.7×
75k	0.0077	0.028	0.296	10.6×
99k	0.0071	0.026	0.292	11.2×

- The **HP1/Baseline ratio stays constant at ~10–11×** from step 0 to step 99k. This ratio never drifts.
- Diffusion consistently has the lowest a0 test loss throughout: 3–4× lower than Baseline.

Training a0 Shows the Same Pattern

Step	Diffusion train_a0	FM Baseline train_a0	HP Tune1 train_a0	HP1 / BL ratio
0–1k	0.041	0.137	1.365	10.0×
25k	0.005	0.012	0.151	12.6×
50k	0.023	0.032	0.206	6.4×
75k	0.035	0.013	0.153	11.8×
99k	0.002	0.013	0.253	19.5×

- Train a0 ratio is noisier (batch-level), but hovers around ~10× on average, consistent with the stable test a0 ratio.
- Both train and test a0 show the same ~10× offset — this is not a generalization issue, the offset is present in both.

a0-to-Total Loss Ratio

Run	a0_test / test_loss (final)
Diffusion	0.0071 / 0.0276 = 0.26
FM Baseline	0.0257 / 0.117 = 0.22
FM HP Tune1	0.292 / 0.064 = 4.56

- For Diffusion and Baseline: a0 is 22–26% of total test loss.
- For HP Tune1: a0 is **4.56×** the total test loss.
- The a0 and test_loss metrics are on **incomparable scales** for HP Tune1 vs the other two runs. Direct a0 comparison between HP Tune1 and the others is not meaningful without understanding the scale difference.

a0 Reduction Factor

Run	a0 Test Start	a0 Test End	Reduction
Diffusion	0.112	0.0071	15.8×
FM Baseline	0.156	0.0257	6.1×
FM HP Tune1	1.562	0.292	5.4×

- Diffusion reduces a0 by 15.8× over training — strongest a0 learning.
- HP Tune1 and Baseline reduce by ~5–6× each — comparable relative learning rates on a0.

6. Summary of Observed Facts



What the data shows (Seed 6):

1. **Test loss ranking:** Diffusion (0.028) < HP Tune1 (0.064) < Baseline (0.117).
2. **a0 test ranking:** Diffusion (0.007) < Baseline (0.026) < HP Tune1 (0.292). Ranking inversion vs test loss for the two FM runs.
3. **Convergence:** Diffusion and HP Tune1 are 5–14× faster than Baseline. Diffusion is 1.4× faster than HP Tune1 at the tightest threshold.
4. **Plateau:** All three models do 85%+ of learning by 10k steps. After 50k, gains are <0.008 for all.
5. **HP Tune1 stall:** HP Tune1 has a flat phase at 30–50k ($\Delta = 0.000$), then resumes 50–75k.
6. **Generalization gap:** Absolute gap varies (0.016 to 0.059), but relative gap (% of test loss) is similar (~51–57%) across all three.
7. **a0 scale:** HP Tune1 a0 is consistently ~10–11× larger than Baseline a0 from step 0 to 99k, in both train and test. The ratio is constant and present from initialization. $a0/test_loss = 0.22\text{--}0.26$ for Diffusion and Baseline, but 4.56 for HP Tune1 — the a0 metric is on a different scale for HP Tune1.
8. **a0 learning:** All three runs reduce a0 over training. Diffusion is most efficient (15.8×). HP Tune1 and Baseline have similar relative a0 improvement (~5–6×).